

SHADRINSK, Russian Federation

WMO: 285520

Lat: 56.067N

Lon: 63.650E

Elev: 89

StdP: 100.26

Time Zone: 5.00 (E05)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-31.2	-28.1	-34.2	0.2	-30.8	-31.2	0.2	-28.1	8.1	-4.6	7.2	-7.0	1.0	270	0.583

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.5	30.9	19.9	29.0	19.2	27.2	18.5	21.6	28.4	20.5	26.7	19.5	25.3	2.9	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.3	14.2	24.4	18.3	13.3	23.5	17.3	12.5	22.5	63.1	28.4	59.5	26.9	56.0	25.6	27.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 7.6	6.6	5.9	DB	-33.8	33.5	3.4	2.2	-36.2	35.1	-38.2	36.3	-40.2	37.5	-42.7	39.1	(4)
(5)			WB	-34.0	23.2	3.5	1.4	-36.5	24.3	-38.5	25.1	-40.5	25.9	-43.0	27.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	3.1	-14.8	-12.9	-4.8	5.0	12.8	17.5	19.0	17.0	11.1	4.1	-5.5	-11.8	(6)	
	DBStd	13.35	7.07	7.12	5.19	5.39	5.16	4.54	3.88	4.16	4.72	4.90	7.30	7.38	(7)	
	HDD10.0	3448	771	641	459	165	31	3	0	2	42	191	466	677	(8)	
	HDD18.3	5701	1029	874	718	399	183	67	41	77	223	440	716	935	(9)	
	CDD10.0	941	0	0	0	16	118	229	278	218	74	8	0	0	(10)	
	CDD18.3	154	0	0	0	0	11	43	60	35	5	0	0	0	(11)	
	CDH23.3	1637	0	0	0	7	167	472	609	327	54	0	0	0	(12)	
	CDH26.7	499	0	0	0	1	38	154	194	100	12	0	0	0	(13)	
(14) Wind	WSAvg	2.5	2.2	2.3	2.7	3.0	2.8	2.4	2.1	2.1	2.3	2.7	2.6	2.4	(14)	
(15) Precipitation	PrecAvg	433	22	17	14	22	39	58	71	59	42	35	28	25	(15)	
	PrecMax	616	66	59	47	58	106	119	143	117	161	96	57	54	(16)	
	PrecMin	251	2	1	0	0	8	12	8	10	8	7	1	0	(17)	
	PrecStd	83	14	12	10	14	23	30	33	27	35	22	13	12	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	1.7	3.4	8.6	23.7	30.2	32.7	33.7	32.5	28.4	19.8	8.8	2.9	(19)	
		MCWB	0.5	1.1	4.6	13.0	16.7	20.1	21.0	20.8	16.7	11.8	6.1	1.3	(20)	
	2%	DB	-0.6	1.4	5.2	19.5	27.3	30.5	30.7	29.2	24.6	16.6	6.4	1.5	(21)	
		MCWB	-1.7	-0.3	2.3	11.2	15.8	19.4	20.7	19.6	15.9	10.5	4.5	0.1	(22)	
	5%	DB	-2.6	-0.7	3.6	16.4	25.0	28.4	28.7	26.9	21.3	13.7	4.6	-0.1	(23)	
		MCWB	-3.6	-2.0	1.3	9.6	15.3	19.0	19.9	19.0	14.8	9.0	2.9	-1.2	(24)	
	10%	DB	-4.7	-2.8	2.2	13.6	22.5	26.0	26.8	24.5	18.8	11.2	2.8	-2.0	(25)	
		MCWB	-5.6	-4.1	0.3	8.3	13.7	18.0	19.2	18.1	13.5	7.9	1.4	-3.0	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	0.8	1.6	4.9	13.9	19.0	22.5	22.8	22.4	18.4	12.9	6.5	1.6	(27)	
		MCDB	1.8	3.1	8.0	23.0	25.2	29.7	29.1	29.4	24.7	17.8	8.2	2.7	(28)	
	2%	WB	-1.7	-0.3	2.7	11.8	16.8	20.9	21.8	20.9	16.5	11.1	4.6	0.3	(29)	
		MCDB	-0.7	1.1	4.8	18.2	25.0	27.3	28.6	27.7	22.9	15.6	6.2	1.4	(30)	
	5%	WB	-3.5	-2.0	1.6	10.2	15.6	19.7	20.8	19.6	15.1	9.4	3.1	-1.2	(31)	
		MCDB	-2.6	-0.6	3.4	15.9	23.4	26.3	27.0	25.1	20.5	12.9	4.4	-0.2	(32)	
	10%	WB	-5.5	-4.1	0.4	8.6	14.4	18.6	19.9	18.5	13.7	8.0	1.5	-3.0	(33)	
		MCDB	-4.7	-2.9	2.1	13.2	21.4	24.8	25.5	23.5	18.1	11.1	2.7	-1.9	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	7.2	8.7	8.6	9.5	11.6	10.8	10.5	10.0	9.3	7.1	5.9	6.5	(35)	
		MCDBR	7.0	7.3	8.2	13.9	15.5	14.3	14.0	13.7	13.4	11.6	6.7	6.5	(36)	
	5% WB	MCWBR	6.6	6.3	6.1	7.4	7.1	5.9	5.4	5.7	6.4	6.8	5.1	5.7	(37)	
		MCWBR	6.9	7.2	7.4	12.8	14.0	12.6	12.2	12.2	12.4	10.8	5.8	6.6	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.287	0.318	0.371	0.415	0.416	0.410	0.430	0.444	0.387	0.356	0.313	0.287	(40)	
		taud	2.048	2.025	1.948	2.089	2.200	2.290	2.231	2.158	2.281	2.238	2.131	2.016	(41)	
	Ebn at Noon		576	712	760	794	823	834	805	760	756	672	543	462	(42)	
		Edh at Noon	64	98	136	138	132	122	127	129	99	80	59	52	(43)	
(44) All-Sky Solar Radiation	RadAvg		0.80	1.84	3.40	4.41	5.26	5.75	5.52	4.28	2.82	1.43	0.71	0.49	(44)	
	RadStd		0.07	0.10	0.20	0.43	0.50	0.55	0.44	0.44	0.25	0.23	0.09	0.06	(45)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (10 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page